Chapter 9 Intro to Networking Exam

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Date: 10/18/2021 7:42:35 pm • Time spent: 18:11

Score: 97% Passing Score: 80%

Question 1: ✔ Correct

Which of the following cloud computing solutions will deliver software applications to a client either over the internet or on a local area network?

- PaaS
- IaaS
- DaaS
- ➡ SaaS

EXPLANATION

Software as a service (SaaS) delivers software applications to the client either over the internet or on a local area network.

Infrastructure as a service (IaaS) delivers infrastructure to the client, such as processing, storage, networks, and virtualized environments. The client deploys and runs software without purchasing servers, data center space, or network equipment. Platform as a service (PaaS) delivers everything a developer needs to build an application onto the cloud infrastructure. The deployment comes without the cost and complexity of buying and managing the underlying hardware and software layers. Data as a service (DaaS) stores and provides data from a centralized location without requiring local collection and storage.

REFERENCES

LabSim for Network Pro, Section 9.5.

Question 2: ✔ Correct

Which of the following does not accurately describe an iSCSI SAN?

- Can authenticate connections and encrypt data transmissions.
- Needs dedicated Ethernet cabling.
- Uses port 3260 by default.
- ➡ Can be implemented on a standard production network with other network traffic.

EXPLANATION

ISCI should never be implemented on a standard production network. The performance of the SAN will be heavily impacted. Best practice is to use dedicated network infrastructure.

REFERENCES

LabSim for Network Pro, Section 9.1.

Question 3: ✔ Correct

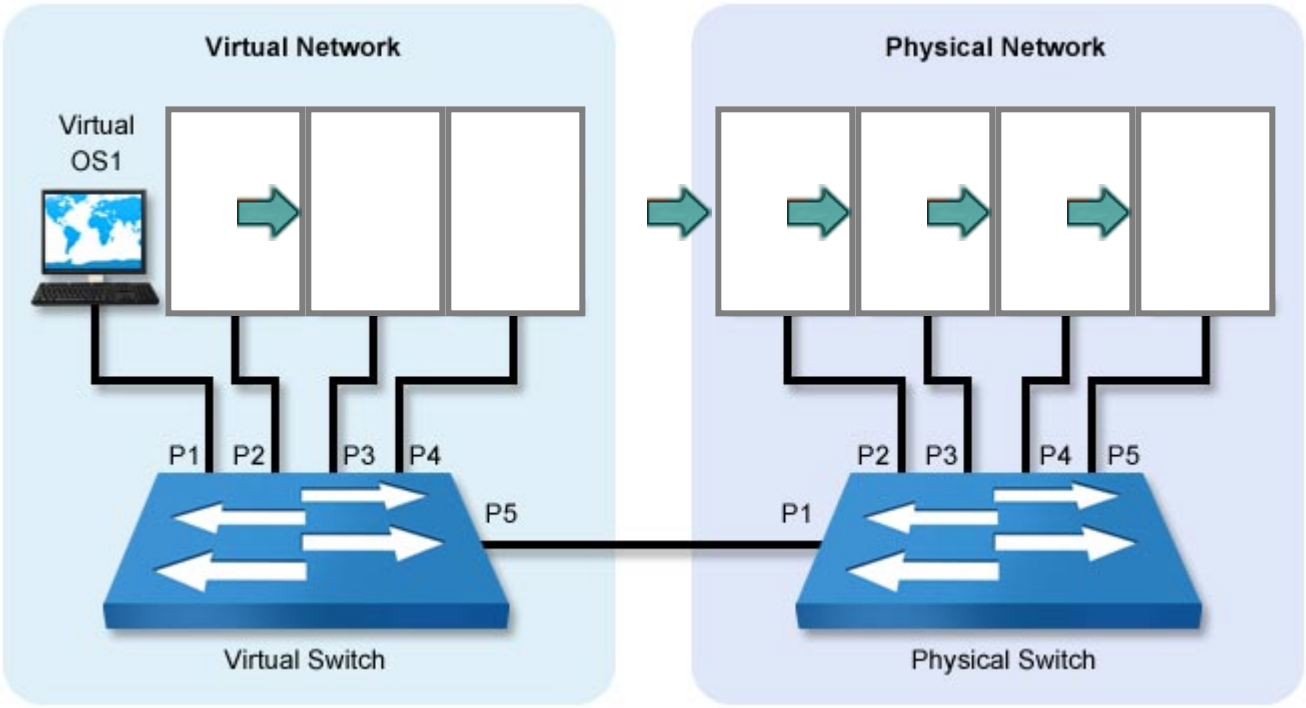
You have configured a virtual network that includes the following virtual components:

- Four virtual machines (Virtual OS1, Virtual OS2, etc.)
- One virtual switch

The virtual switch is connected to a physical network to allow the virtual machines to communicate with the physical machines out on the physical network.

Given the port configuration for the virtual switch and the physical switch in the table below, click on all of the virtual and physical machines that Virtual OS1 can communicate with.

Device	Port	Port Assignment
Virtual Switch	P1	Virtual Network1
	P2	Virtual Network2
	P3	Virtual Network1
	P4	Virtual Network2
	P5	Physical Network, Virtual Network1
Physical Switch	P1	Physical Network
	P2	Physical Network
	P3	Physical Network
	P4	Physical Network
	P5	Physical Network



EXPLANATION

Virtual OS1 can communicate with the following machines:

- Virtual OS3
- Physical OS1
- Physical OS2
- Physical OS3
- Physical OS4

The virtual switch port configuration allows Virtual OS1 to communicate with machines on Virtual Network1 and the physical network. P5 on the virtual switch is configured to allow communication between the virtual and physical machines as if they were on the same real physical network.

Virtualized networks allow virtual servers and desktops to communicate with each other, and can also allow communication with network devices out on the physical network via the host operating system. Virtual networks typically include the following components:

- Virtual switches allow multiple virtual servers and/or desktops to communicate on virtual network segments and/or the physical network. Virtual switches are often configured in the hypervisor.
- Virtual network adapters are created and assigned to a desktop or server in the hypervisor.
  - Multiple network adapters could be assigned to a single virtual machine.
  - Each network adapter has its own MAC address.
  - Each network adapter is configured to connect to only one network at a time (meaning a virtual network or the physical network, but not both).

Virtual OS2 and Virtual OS4 belong to Virtual Network2 and are only able to communicate with each other.

REFERENCES

LabSim for Network Pro, Section 9.4.

Question 4: ✔ Correct

You are in the process of configuring an iSCSI storage area network (SAN) for your network.

You want to configure a Windows Server 2016 system to connect to an iSCSI target defined on a different server system. You also need to define iSCSI security settings, including CHAP and IPsec.

Which tool should you use?

- iSCSI under File and Storage Services in Server Manager
- ➡ iSCSI Initiator
- Multipath I/O
- Internet Storage Name Service

EXPLANATION

Run the iSCSI Initiator to connect to an iSCSI target defined somewhere on the SAN fabric. You can also use this utility to define iSCSI security settings, including CHAP and IPsec.

Internet Storage Name Service (iSNS) servers provide discoverability and zoning of SAN resources. Multipath I/O (MPIO) provides support for using multiple data paths to a storage device. Use the iSCSI option under File and Storage Services in Server Manager to define an iSCSI target on a server.

REFERENCES

LabSim for Network Pro, Section 9.1.

Question 5: ✔ Correct

Which of the following statements about virtual NICs are true? (Select two.)

- [ ] Virtual NICs don't have a MAC address.
- [x] Multiple virtual NICs can be added to a virtual machine.
- [ ] The type of NIC installed in the physical machine determines the type of virtual NIC that is emulated.
- [x] Virtual NICs need the appropriate driver installed to function.
- [ ] Virtual NICs can only communicate with other virtual NICs.

EXPLANATION

Within each virtual machine, you can configure one or more virtual network interfaces, which function similarly to physical network interfaces. Virtual interfaces use Ethernet standards to transmit and receive frames on the network. The operating system within the virtual machine must have the appropriate driver installed to support the virtual network interface, just as with a physical network interface.

When you configure a virtual network interface within a virtual machine's configuration, you can specify:

- The type of physical network interface to emulate. This allows the best possible driver support from the operating system within the virtual machine.
- A MAC address. Most hypervisors automatically assign a MAC address to each virtual network interface. Some hypervisors allow you to use a custom MAC address if needed.
- The network to connect to. Most hypervisors allow you to define many different virtual networks.
- When you configure a virtual network interface, you will select which virtual network you want it to be connected to.

REFERENCES

LabSim for Network Pro, Section 9.4.

Question 6: ✔ Correct

How can QoS be configured so that large data transfers will not block VoIP calls by using too much network bandwidth?

- ➡ QoS can be configured on network devices to give priority to VoIP traffic.
- QoS can be configured on network devices to limit the size of a file that can be transferred on the network.
- QoS can be configured on network devices to only allow network protocols that throttle network bandwidth usage.
- QoS can be configured on network devices to set a bandwidth threshold on selected ports.

EXPLANATION

Network devices can examine the type of service or precedence bits in the header of an IP packet to determine the type of traffic. QoS settings can be configured on a network devices to give VoIP traffic priority over normal computer traffic.

REFERENCES

LabSim for Network Pro, Section 9.2.

Question 7: ✔ Correct

You need to provide DHCP and file share services to a physical network. These services should be deployed using virtualization. Which type of virtualization should you implement?

- ➡ Virtual servers
- Network as a service (NaaS)
- Virtual desktops
- Virtual networks

EXPLANATION

Server virtualization runs multiple instances of a server operating system on a single physical computer. With server virtualization, you can migrate servers on older hardware to newer computers or add virtual servers to computers with extra unused hardware resources.

Virtual desktops do not provide DHCP services. Virtual networks allow virtual servers and desktops to communicate with each other, and they can also allow communication with network devices out on the physical network via the host operating system. Network as a service (NaaS) servers and desktops that are all virtualized and managed by a contracted third party.

REFERENCES

LabSim for Network Pro, Section 9.4.

Question 8: ✔ Correct

You are an application developer. You use a hypervisor with multiple virtual machines installed to test your applications on various operating system versions and editions.

Currently, all of your testing virtual machines are connected to the production network through the hypervisor's network interface. You are concerned that the latest application you are working on could adversely impact other network hosts if errors exist in the code.

To prevent problems, you decide to isolate the virtual machines from the production network. However, they still need to be able to communicate directly with each other.

What should you do? (Select two. Each response is one part of the complete solution.)

- [ ] Disable the switch port that the hypervisor's network interface is connected to.
- [x] Connect the virtual network interfaces in the virtual machines to the virtual switch.
- [ ] Create a new virtual switch configured for bridged (external) networking.
- [x] Create a new virtual switch configured for host-only (internal) networking.
- [ ] Create MAC address filters on the network switch that block each virtual machine's virtual network interfaces.
- [ ] Disconnect the network cable from the hypervisor's network interface.

EXPLANATION

To allow the virtual machines to communicate with each other while isolating them from the production network, complete the following:

- Create a new virtual switch configured for host-only (internal) networking.
- Connect the virtual network interfaces in the virtual machines to the virtual switch.

Creating a bridged virtual switch would still allow the virtual machines to communicate on the production network through the hypervisor's network interface. Disconnecting the hypervisor's network cable, blocking the virtual machine's MAC addresses, or disabling the hypervisor's switch port would isolate the virtual machines from the production network, but would also prevent them from communicating with each other.

REFERENCES

LabSim for Network Pro, Section 9.4.

Question 9: ✔ Correct

What is a soft phone?

- A traditional or VoIP phone that has a padded handset that is more comfortable for a user.
- A software algorithm that compresses VoIP data prior to transmission on an IP network.
- A device that converts fax calls and other soft calls between the PSTN and an IP network.
- ➡ A software application that runs on a computer or other device that accesses a VoIP server to make real-time phone calls.

EXPLANATION

A soft phone is a software application that is installed on a computing device such as a computer or a handheld device.

REFERENCES

LabSim for Network Pro, Section 9.2.

Question 10: ✔ Correct

Arrange the Fibre Channel (FC) SAN implementation tasks in the order they should be performed to build a redundant FC SAN.

Step 1

✔ Install two Fiber Channel adapters in each server.

Step 2

✔ Deploy two FC switches.

Step 3

✔ Using fiber optic access cables, connect each server to each FC switch.

Step 4

✔ Deploy the shared storage devices.

✔ Using fiber optic connecting cables, connect each storage device to each FC switch.

EXPLANATION

A switched FC SAN uses fiber optic cabling, FC host bus adapters (HBAs), and FC switches to build the SAN fabric. To build a redundant Fibre Channel SAN, you need to complete the following:

1. Install two FC HBAs in each server that will access the shared storage on the SAN.
2. Deploy two FC switches.
3. Using fiber optic cables, connect each server to each FC switch.
4. Deploy shared storage devices. Typically, these are external RAID device containing multiple hard disk drives and two FC HBAs.
5. Using fiber optic cables, connect each storage device to each FC switch.

REFERENCES

LabSim for Network Pro, Section 9.1.

Question 11: ✔ Correct

Which of the following are typical components of a NAS device? (Select two.)

- [x] A minimal network OS
- [ ] An FC switch
- [ ] Initiator server
- [x] One or more NICs
- [ ] A dedicated network

EXPLANATION

A NAS device typically consists of:

- A RAID array with terabytes of storage space.
- A motherboard with a processor and memory.
- One or more NICs.
- A minimal network operating system.

REFERENCES

LabSim for Network Pro, Section 9.1.

Question 12: ✔ Correct

To answer this question, complete the lab using the information below.
You have already answered this question.
You are not allowed to view the lab again.

Launch Lab

You completed the lab correctly.

View Lab Report

Question 13: ✔ Correct

You are on a phone call using VoIP. You notice that it takes several seconds for the person on the other end to respond to questions you ask.

Which type of problem is occurring?

- ➡ Latency
- Jitter
- Packet loss
- Echo

EXPLANATION

Because VoIP transmits call data using IP packets over a packet-switched network, VoIP is susceptible to the following problems:

- Delay (or latency), which occurs when data takes a long time to arrive at the receiving device. Delays cause long pauses between speaking and receiving and can result in callers continually interrupting each other. International standards call for a delay of 150 milliseconds or less.
- Jitter, which is a variation in the delay of individual packets. Jitter causes strange sound effects as the delay of packets fluctuates.
- Packet loss, which occurs when packets do not arrive at all. Packet loss causes dropouts in the conversation.
- Echo, which occurs when you hear your own voice in the telephone receiver while you are talking. Excessive delay can cause unacceptable levels of echo.

REFERENCES

LabSim for Network Pro, Section 9.2.

Question 14: ✔ Correct

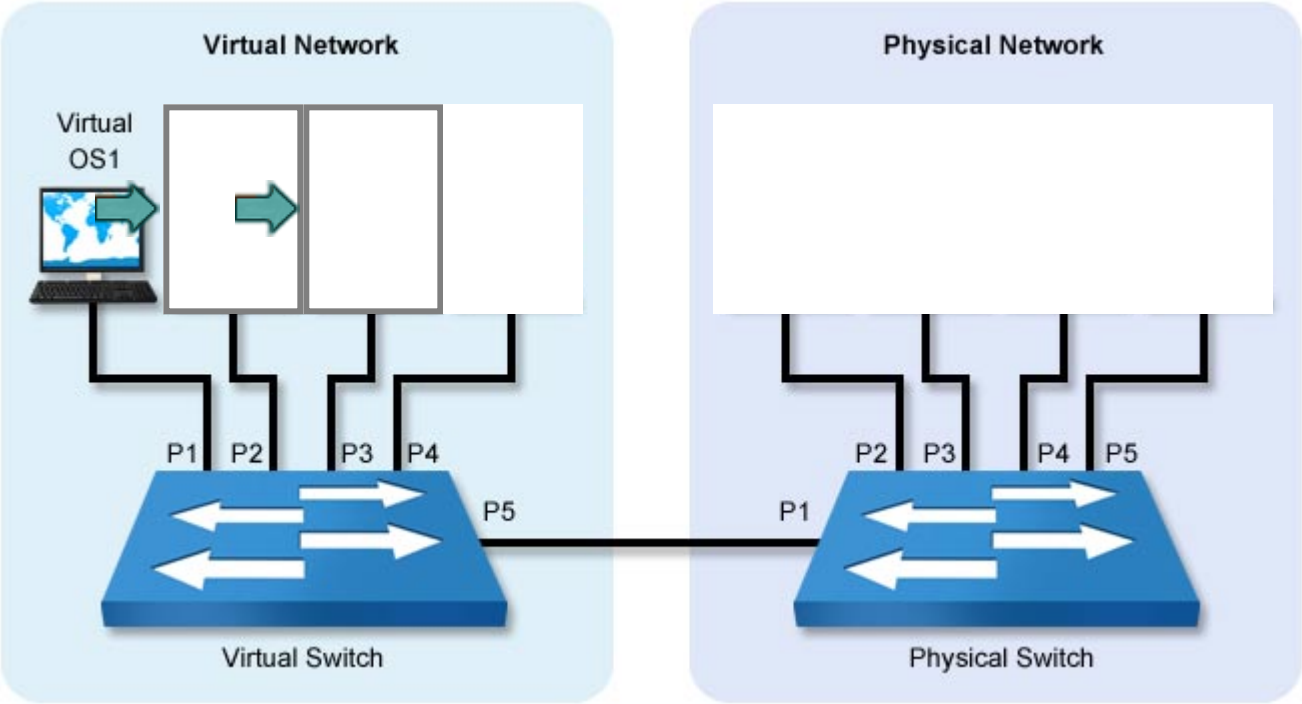
You have configured a virtual network that includes the following virtual components:

- Four virtual machines (Virtual OS1, Virtual OS2, etc.)
- One virtual switch

The virtual switch is connected to a physical network to allow the virtual machines to communicate with the physical machines out on the physical network.

Given the port configuration for the virtual switch and the physical switch in the table below, click on all of the virtual and physical machines that Virtual OS1 can communicate with.

Device	Port	Port Assignment
Virtual Switch	P1	Virtual Network1
	P2	Virtual Network1
	P3	Virtual Network1
	P4	Physical Network
	P5	Physical Network
Physical Switch	P1	Physical Network
	P2	Physical Network
	P3	Physical Network
	P4	Physical Network
	P5	Physical Network



EXPLANATION

Virtual OS1 can communicate with the following machines:

- Virtual OS2
- Virtual OS3

The virtual switch port configuration allows these three virtual machines to communicate as if the machines were part of a real physical network. Virtualized networks allow virtual servers and desktops to communicate with each other and can also allow communication with network devices out on the physical network via the host operating system. Virtual networks typically include the following components:

- Virtual switches allow multiple virtual servers and/or desktops to communicate on virtual network segments and/or the physical network. Virtual switches are often configured in the hypervisor.
- Virtual network adapters are created and assigned to a desktop or server in the hypervisor.
  - Multiple network adapters can be assigned to a single virtual machine.
  - Each network adapter has its own MAC address.
  - Each network adapter is configured to connect to only one network at a time (meaning a virtual network or the physical network, but not both).

Virtual OS4 and all of the other Physical OS machines are configured to communicate on the physical network.

REFERENCES

LabSim for Network Pro, Section 9.4.

Question 15: ✔ Correct

Which component is most likely to allow physical and virtual machines to communicate with each other?

- ➡ Virtual switch
- Virtual desktop
- VHD
- Host operating system

EXPLANATION

Virtual switches allow multiple virtual servers and/or desktops to communicate on virtual network segments and/or the physical network. Virtual switches are often configured in the hypervisor.

A virtual hard disk (VHD) is a file that is created within the host operating system and simulates a hard disk for the virtual machine. A physical machine (also known as the host operating system) has the actual hardware in place on the machine, such as the hard disk drive(s), optical drive, RAM, motherboard, etc. A virtual desktop is a virtual machine in a software implementation of a computer that executes programs like a physical machine.

REFERENCES

LabSim for Network Pro, Section 9.3.

Question 16: ✔ Correct

Which of the following are advantages of virtualization? (Select two.)

- [ ] Reduced utilization of hardware resources
- [x] Easy system migration to different hardware
- [ ] Redundancy of hardware components for fault tolerance
- [ ] Improved detection of host-based attacks
- [x] Centralized administration

EXPLANATION

Virtualization allows a single physical machine (known as the host operating system) to run multiple virtual machines (known as the guest operating systems). The virtual machines appear to be self-contained and autonomous systems. Advantages of virtualization include:

- Server consolidation
- The ability to migrate systems between different hardware
- Centralized management of multiple systems
- Increased utilization of hardware resources
- Isolation of systems and applications

Disadvantages of virtualization include:

- A compromise in the host system could affect multiple guest systems.
- A failure in a shared hardware resource could affect multiple systems.

REFERENCES

LabSim for Network Pro, Section 9.3.

Question 17: ✔ Correct

Which VoIP device helps establish the connection between two VoIP phones?

- VoIP endpoint
- VoIP gateway
- ➡ VoIP server
- VoIP codec

EXPLANATION

A VoIP server helps establish the connection between two VoIP phones. Once the connection is established, the two phones communicate directly with each other.

A VoIP gateway converts voice calls between the PSTN and an IP network. A VoIP endpoint is a client of a VoIP server, such as a VoIP phone. A VoIP codec is not a device, but a special algorithm that compresses VoIP data to reduce bandwidth consumption.

REFERENCES

LabSim for Network Pro, Section 9.2.

Question 18: ✔ Correct

Upper management has asked you if there is a way to integrate phone calls, emails, and instant messaging into a single platform.

Which of the following systems should you recommend?

- PSTN
- Voice over IP
- ➡ Unified communication
- Quality of service

EXPLANATION

Unified communications (UC) integrates multiple types of communications into a single system. UC systems can integrate the following real-time communications:

- Voice calls
- Audio conferencing
- Video conferencing (VTC)
- Desktop sharing
- Instant messaging

UC systems can also provide non-real-time communication integration, including:

- Texting
- Voicemail
- Email
- Faxing

Voice over IP only provides voice calling integration with an IP network. Quality of Service is used to ensure that voice data is given higher priority on the network. The PSTN is the traditional method used for phone calls.

REFERENCES

LabSim for Network Pro, Section 9.2.

Question 19: ✔ Correct

In virtualization, what is the role of the hypervisor?

- A hypervisor is a software implementation that executes programs like a physical machine.
- ➡ A hypervisor allows virtual machines to interact with the hardware without going through the host operating system.
- A hypervisor has the actual hardware in place on the machine, such as the hard disk drive(s), optical drive, RAM, and motherboard.
- A hypervisor is created within the host operating system and simulates a hard disk for the virtual machine.

EXPLANATION

A hypervisor is a thin layer of software that resides between the virtual operating system(s) and the hardware. A hypervisor allows virtual machines to interact with the hardware without going through the host operating system. A hypervisor manages access to system resources such as:

- CPU
- Storage
- RAM

A physical machine (also known as the host operating system) has the actual hardware in place on the machine, such as the hard disk drive(s), optical drive, RAM, motherboard, etc. A virtual machine is a software implementation that executes programs like a physical machine. A virtual machine appears to be a self-contained and autonomous system. A virtual hard disk (VHD) is a file that is created within the host operating system and simulates a hard disk for the virtual machine.

REFERENCES

LabSim for Network Pro, Section 9.3.

Question 20: ✔ Correct

What is one benefit of placing VoIP gateways in geographically separated branch offices that have an existing WAN connection?

- Data can be transported between VoIP gateways over the PSTN instead of via the expensive WAN.
- VoIP gateways can aggregate multiple VoIP calls for more efficient transmission to the home office.
- Less costly VoIP gateways can be used in place of expensive routers and modems to provide WAN connectivity.
- ➡ Long-distance PSTN charges can be reduced by switching VoIP calls to the PSTN in locations where only local call charges would be incurred.

EXPLANATION

VoIP gateways convert voice and fax calls between the PSTN and an IP network. A WAN connection can carry VoIP calls from a distant location to a geographically separated branch office. A VoIP gateway located at the branch office can switch the call to the PSTN, where only local phone charges would be incurred.

REFERENCES

LabSim for Network Pro, Section 9.2.

Question 21: ✔ Correct

You are responsible for maintaining Windows workstation operating systems in your organization. Recently, an update from Microsoft was automatically installed on your workstations that caused an in-house application to stop working.

To keep this from happening again, you decide to test all updates on a virtual machine before allowing them to be installed on production workstations.

Currently, none of your testing virtual machines has a network connection. However, they need to be able to connect to the update servers at Microsoft to download and install updates.

What should you do? (Select two. Each response is one part of the complete solution.)

- [x] Create a new virtual switch configured for bridged (external) networking.
- [ ] Create a new virtual switch configured for host-only networking.
- [ ] Disable the switch port that the hypervisor's network interface is connected to.
- [ ] Create a new virtual switch configured for internal networking.
- [x] Connect the virtual network interfaces in the virtual machines to the virtual switch.

EXPLANATION

To allow the virtual machines to communicate with the Microsoft update servers on the internet, complete the following:

- Create a new virtual switch configured for bridged (external) networking.
- Connect the virtual network interfaces in the virtual machines to the virtual switch.

Creating an internal or host-only virtual switch would not allow the virtual machines to communicate on the production network through the hypervisor's network interface. Disabling the hypervisor's switch port would also isolate the virtual machines from the production network.

REFERENCES

LabSim for Network Pro, Section 9.4.

Question 22: ✔ Correct

Which of the following protocols is used by VoIP to set up, maintain, and terminate a phone call?

- TLS
- ➡ SIP
- NTP
- SSH
- RTP

EXPLANATION

The session initiation protocol (SIP) is used to set up, maintain, tear down, and redirect the call. The real-time transport protocol (RTP) contains the actual voice data.

SSH is used for secure remote administration of a network device. TLS is used to add security to other protocols. NTP is used for synchronizing clocks on network devices.

REFERENCES

LabSim for Network Pro, Section 9.2.

Question 23: ✔ Correct

Which switch features are typically used with VoIP? (Select two.)

- [x] VLAN
- [ ] Mirroring
- [ ] Spanning tree
- [x] PoE

EXPLANATION

When configuring Voice over IP (VoIP), switches with Power over Ethernet (PoE) capabilities provide power to the VoIP phone through an Ethernet cable, the same cable that is used for transmitting data signals. Virtual LANs (VLANs) are often used to distinguish voice traffic from data traffic so that Quality of Service (QoS) measures can be applied to traffic that is part of the voice VLAN.

Bonding allows multiple switch ports to be used at the same time to reach a specific destination. Spanning tree is a protocol on a switch that allows the switch to maintain multiple paths between switches within a subnet. The spanning tree protocol runs on each switch and is used to select a single path between any two switches. Mirroring sends traffic from all switch ports to a switch port you designate as the mirrored port.

REFERENCES

LabSim for Network Pro, Section 9.2.

Question 24: ✔ Correct

Match each description on the left with the appropriate cloud technology on the right.

Public cloud	Private cloud	Community cloud
✔ Provides cloud services to just anyone.	✔ Provides cloud services to a single organization.	✔ Allows cloud services to be shared by several organizations.

Hybrid cloud

✔ Integrates one cloud service with other cloud services.

EXPLANATION

Cloud computing can be implemented in several different ways, including the following:

- A public cloud can be accessed by anyone. Cloud-based computing resources are made available to the general public by a cloud service provider. The service provider may or may not require a fee for using these resources. For example, Google provides many publicly accessible cloud applications, such as Gmail and Google Docs.
- A private cloud provides resources to a single organization. Access is restricted to only the users within that organization. An organization commonly enters into an agreement with a cloud service provider that provides secure access to cloud-based resources. The organization's data is kept separate and secure from any other organization using the same service provider.
- A community cloud is designed to be shared by several organizations. Access is restricted to only users within the organizations who are sharing the community cloud infrastructure. Community clouds are commonly hosted externally by a third party.
- A hybrid cloud is composed of a combination of public, private, and community cloud resources from different service providers. The goal behind a hybrid cloud is to expand the functionality of a given cloud service by integrating it with other cloud services.

REFERENCES

LabSim for Network Pro, Section 9.5.

Question 25: ✕ Incorrect

To answer this question, complete the lab using the information below. **Launch Lab**

You did not attempt the lab.

Question 26: ✔ Correct

Which of the following features is used with digital IP phones to supply power through a switch port?

- ➡ PoE
- Spanning tree
- VPN
- 802.1x
- Trunking

EXPLANATION

Power over Ethernet (PoE) supplies power to end devices through the RJ45 Ethernet switch port. Power to the phone is carried on unused wires within the drop cables.

Spanning tree is a protocol on a switch that allows the switch to maintain multiple paths between switches within a subnet. The spanning tree protocol runs on each switch and is used to select a single path between any two switches. Trunking allows a switch to forward VLAN traffic between switches. 802.1x is an authentication protocol used with port security or port authentication.

REFERENCES

LabSim for Network Pro, Section 9.2.

Question 27: ✔ Correct

Which of the following are true regarding cloud computing? (Select three.)

- [x] Cloud computing is software, data access, computation, and storage services provided to clients through the internet.
- [x] The term "cloud" is used as a synonym for the internet.
- [x] Typical cloud computing providers deliver common business applications online that are accessed from another web service or software like a web browser.
- [ ] Cloud computing requires end user knowledge of the physical location and configuration of the system that delivers the services.

EXPLANATION

Cloud computing does not require end user knowledge of the physical location and configuration of the system that delivers the services. Other cloud computing details include the following:

- Cloud computing is software, data access, computation, and storage services provided to clients through the internet.
- The term "cloud "is used as a synonym for the internet based on the basic cloud drawing used to represent the telephone network infrastructure (and, later, the internet) in computer network diagrams.
- Typical cloud computing providers deliver common business applications online that are accessed from another web service or software like a web browser, while the software and data are stored on servers.

REFERENCES

LabSim for Network Pro, Section 9.5.

Question 28: ✔ Correct

Which of the following protocols is an open source protocol used by most manufacturers of VoIP systems?

- User datagram protocol (UDP)
- ➡ Session initiation protocol (SIP)
- Transmission control protocol (TCP)
- Stream control transmission protocol (SCTP)

EXPLANATION

The session initiation protocol (SIP) is one of the protocols used during the call control process of multimedia communications, such as a VoIP call. SIP is used to set up, maintain, and tear down multimedia communications.

SIP rides on top of the TCP, UDP, and SCTP transport layer protocols.

REFERENCES

LabSim for Network Pro, Section 9.2.

Question 29: ✔ Correct

Your company uses VoIP for phone calls. Recently, employees have been complaining about phone calls with unusual sound effects.

Which type of problem is occurring on the VoIP system?

- Latency
- Packet loss
- ➡ Jitter
- Echo

EXPLANATION

Because VoIP transmits call data using IP packets over a packet-switched network, VoIP is susceptible to the following problems:

- Latency occurs when data takes a long time to arrive at the receiving device. Delays cause long pauses between speaking and receiving and can result in callers continually interrupting each other.
- Jitter is a variation in the delay of individual packets. Jitter causes strange sound effects as the delay of packets fluctuates.
- Packet loss occurs when packets do not arrive at all. Packet loss causes drop-outs in the conversation.
- Echo occurs when you hear your own voice in the telephone receiver while you are talking. Excessive delay can cause unacceptable levels of echo.

REFERENCES

LabSim for Network Pro, Section 9.2.

Question 30: ✔ Correct

What are other names for a VoIP server? (Select two.)

- [ ] Jitter
- [ ] QoS
- [x] IP-PBX
- [ ] Hard phone
- [x] VoIP PBX

EXPLANATION

A VoIP server is also known as a VoIP PBX or an IP-PBX since a VoIP server provides many of the functions of a traditional phone system PBX.

Quality of service (QoS) can be configured on network devices to give priority to VoIP traffic. Jitter can cause unusual sound effects in a VoIP call. A hard phone is a VoIP endpoint that is really a computer built to look like and work like a phone.

REFERENCES

LabSim for Network Pro, Section 9.2.

Question 31: ✔ Correct

This question includes an image to help you answer the question. **View Image**

You manage a network with three dedicated storage devices, as shown in the diagram. Users on the network see only a single file server.

Which network-based storage technology is being used?

- Fibre channel SAN
- NAS
- ➡ NAS with clustering
- iSCSI SAN with clustering

EXPLANATION

NAS with clustering is being used. A NAS device is an appliance that is dedicated to file storage. With clustering, multiple NAS devices are grouped together to provide a degree of fault tolerance. To users on the network, the cluster appears as a single file server. Without clustering, the NAS devices would appear as three separate file servers.

Because client devices are connected directly to the switch, it cannot be an iSCSI or Fiber Channel SAN implementation. iSCSI and Fibre Channel SANs both use special switches to create the SAN fabric that client systems are not connected to directly.

REFERENCES

LabSim for Network Pro, Section 9.1.

Question 32: ✔ Correct

When would you consider changing the codec used in your VoIP system? (Select two.)

- [x] When VoIP data consumes too large a portion of your network bandwidth.
- [ ] When an open source VoIP protocol requires a different codec.
- [x] When sound quality is poor.
- [ ] When excessive jitter causes unusual sound effects in VoIP calls.
- [ ] When network latency causes callers to talk over each other.

EXPLANATION

A special algorithm called a codec compresses VoIP data to reduce bandwidth consumption. A codec determines the sound quality of the VoIP call and the amount of bandwidth that it will require.

A more efficient codec can better compress VoIP data so that it will not consume as much bandwidth. Often, better compression will reduce the sound quality of VoIP calls. Conversely, a codec can preserve sound quality at the cost of using more bandwidth. A good codec can preserve sound quality and reduce bandwidth consumption.

REFERENCES

LabSim for Network Pro, Section 9.2.

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